Boxuan Wang

Design Engineering Student

Robotics | Intelligent Systems | jw4424@ic.ac.uk | mystify59.github.io

PROFILE

Design Engineering student at Imperial College London with experience across product design, humanoid development, robot testing and Python. I combine creative problem-solving with technical analysis, CAD and hands-on prototyping to develop practical solutions that address both user needs and engineering challenges.

EDUCATION

Imperial College London 2024 - 2028

Design Engineering | Expected completion 2028

Relevant study: Intelligent Systems, Physical Computing, Mechanical Engineering, FEA.

The King's School, Canterbury 2019 - 2024

King's Scholar; Vice Captain of Fencing.

WORK EXPERIENCE

Li Auto, *Humanoid Development Intern* July - September 2026

- Collected human-motion data using wearable trackers, VR headsets and robot arms mirroring operators for tasks including parts organisation.
- Designed connector parts and structural supports in SolidWorks, supported by Ansys analysis.
- Redesigned upper-torso load paths as part of the team, reducing torso weight by **18%** and meeting fatigue requirements in Ansys simulations with a safety factor of 3.

Elibot, *Cooperative Robot Intern* July - August 2023

- Created joint-specific fatigue tests, logged faults and examined angular differences during braking in harmonic-drive joints.
- Built a Python system to log incoming, produced and shipped robots.
- Contributed to app and web interfaces for remote robot control and webpage UX.

SELECTED PROJECTS

Autonomous Football Robot, *Team robotics project* 2025

- Contributed to team model training and ball-tracking code using OpenMV and Edge Impulse.
- Contributed to team control-code development for autonomous navigation and task execution on an existing robot platform.

Street Pulse, *Performance project* 2025

- Conducted interviews and observations with street performers and audience members to understand engagement, performer needs and opportunities for an interactive-floor product.
- Used sketches, renders and 3D printed lo-fi prototypes in co-design workshops to explore street-performance scenarios and refine the modules' shape, tessellation and arrangement.
- Prototyped electronics, assembled and tested a vibration module, translating research insights into a physical prototype within the team project.

SUV Aerodynamics and Battery Cooling, *Thermofluids project* 2026

- Built SUV geometry in Fusion and a 400,000-cell Ansys Fluent model to analyse pressure distribution and rear-wake behaviour.
- Contributed to a team cold-plate battery-cooling design, evaluating flow distribution, heat transfer and thermal-performance trade-offs.

SKILLS

Programming and vision. Python, C++, MATLAB; OpenMV, Edge Impulse.

Mechanical design. SolidWorks, Fusion; CAD, design for additive manufacturing, prototype assembly.

Analysis and testing. Ansys Mechanical, Fluent; FEA, CFD, fatigue analysis and robot-joint testing.

AI tools. AI agents and MCP tools for 3D-printable validation structures and temporary prototypes.

LANGUAGES

Chinese (Native) | **English** (Fluent) | **French** (Basic)